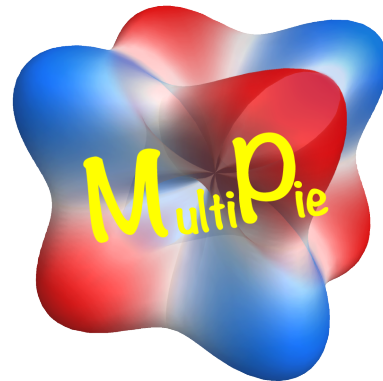


# MultiPie Manual



<https://cmt-mu.github.io/MultiPie/>

for Version 2.2 (2026.7)

Hiroaki Kusunose

## Features

### • Group-theoretical data

| content            | point group | space group | magnetic PG | magnetic SG |
|--------------------|-------------|-------------|-------------|-------------|
| symmetry operation | ✓           | ✓           | ✓           | ✓           |
| Wyckoff (site)     | ✓           | ✓           | ✓           | ✓           |
| Wyckoff (bond)     | ✓           | ✓           |             |             |
| character          | ✓           |             |             |             |
| harmonics          | ✓           |             |             |             |
| active multipole   |             |             | ✓           |             |

### • Symmetry Adapted Multipole Basis (SAMB) data

| content                       | point group | space group | magnetic PG | magnetic SG |
|-------------------------------|-------------|-------------|-------------|-------------|
| PG Clebsch-Gordan coefficient | ✓           |             |             |             |
| atomic SAMB (X)               | ✓           |             |             |             |
| site/bond-cluster SAMB (Y)    | ✓           | ✓           |             |             |
| combined SAMB (Z)             | ✓           | ✓           |             |             |

### • List of group-theoretical data (PDF)

[https://cmt-mu.github.io/MultiPie/src/group\\_doc.html](https://cmt-mu.github.io/MultiPie/src/group_doc.html)

### • Model construction and physical quantity calculations using SAMB

\$ mp\_create ./input\_file.py      Generate SAMB from input model file

\$ mp\_analyze ./control\_file.py    Compute physical quantities from control file

# MultiPie - Symmetry Adapted Multipole Basis (SAMB)

Physical degrees of freedom (classified by the eigen values of the following basic symmetry operations)

- Time-reversal parity (electric or magnetic)
- Spatial-inversion property (polar or axial)
- Spatial anisotropy (rank and component)

| Time/Space | polar                 | axial                 |
|------------|-----------------------|-----------------------|
| even       | Q (electric)          | G (electric toroidal) |
| odd        | T (magnetic toroidal) | M (magnetic)          |

$$X_{lm} (X = Q, G, T, M)$$

Point groups are subgroups of the rotation group

Unitary trans.

classified by PG irreps.  
( $\Gamma, n$ ): irrep. and multiplicity  
 $\gamma$ : component

$$X_{lm} \rightarrow X_{l\Gamma n\gamma} = \sum_m c_{\Gamma n\gamma}^{lm} X_{lm}$$

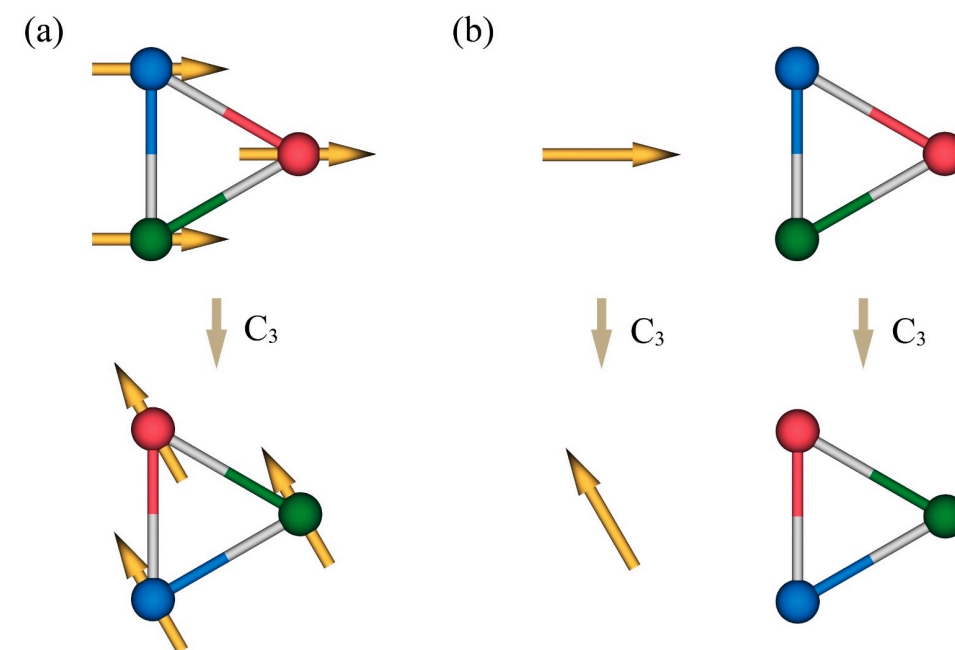
PG symmetry op. acts independently to internal d.o.f. and structure

- internal d.o.f. (atomic multipole)  $X_{lm}$
- structure (cluster multipole)  $Y_{lm}$
- combined basis (combined multipole)

$$Z_{lm} = \sum_{l_1 m_1, l_2 m_2} c_{lm}^{l_1 m_1, l_2 m_2} X_{l_1 m_1} Y_{l_2 m_2}$$



$$\Rightarrow \{ Z_j \}$$



symmetry  
operation

internal d.o.f.  
(atomic)

structure  
(cluster)

# MultiPie - Model Construction (Example : Graphene)

## Input file

```
# graphene_in.py
graphene_in = {
    "model": "graphene", # name of model.
    "group": 191, # No. of space group.
    "cell": {"c": 4}, # set large enough interlayer distance.
    #
    "site": {"C": ("[1/3,2/3,0]", "pz")}, # positions of C site and its orbital.
    "bond": [{"C", "C", [1, 2]}], # C-C bonds up to 2nd neighbors.
    #
    "spinful": False, # spinless.
    #
    "SAMB_select": {"X": []}, # Q, G, T, M (all types)
}
```

**Construction** \$ mp\_create ./graphene\_in.py

**Model output** ./graphene/graphene.pdf

Table 2: Harmonics **symmetry of SAMB**

| # | symbol            | irrep.   | rank | X      | multiplicity | component | symmetry                         |
|---|-------------------|----------|------|--------|--------------|-----------|----------------------------------|
| 1 | $Q_0(A_{1g})$     | $A_{1g}$ | 0    | $Q, T$ | -            | -         | 1                                |
| 2 | $G_1(A_{2g})$     | $A_{2g}$ | 1    | $G, M$ | -            | -         | $z$                              |
| 3 | $Q_3(B_{1u})$     | $B_{1u}$ | 3    | $Q, T$ | -            | -         | $\frac{\sqrt{10}y(3x^2-y^2)}{4}$ |
| 4 | $Q_3(B_{2u})$     | $B_{2u}$ | 3    | $Q, T$ | -            | -         | $\frac{\sqrt{10}x(x^2-3y^2)}{4}$ |
| 5 | $Q_{1,1}(E_{1u})$ | $E_{1u}$ | 1    | $Q, T$ | -            | 1         | $x$                              |
| 6 | $Q_{1,2}(E_{1u})$ |          |      |        |              | 2         | $y$                              |

continued ...

rank, component  
(irrep.)

anisotropy

## Structure of model `./graphene/graphene.qtdw`

C site (representative)  
 $p_z$  orbital only

C-C hopping  
up to 1st and second neighbors

arrow represents  
definition of positive  
bond direction

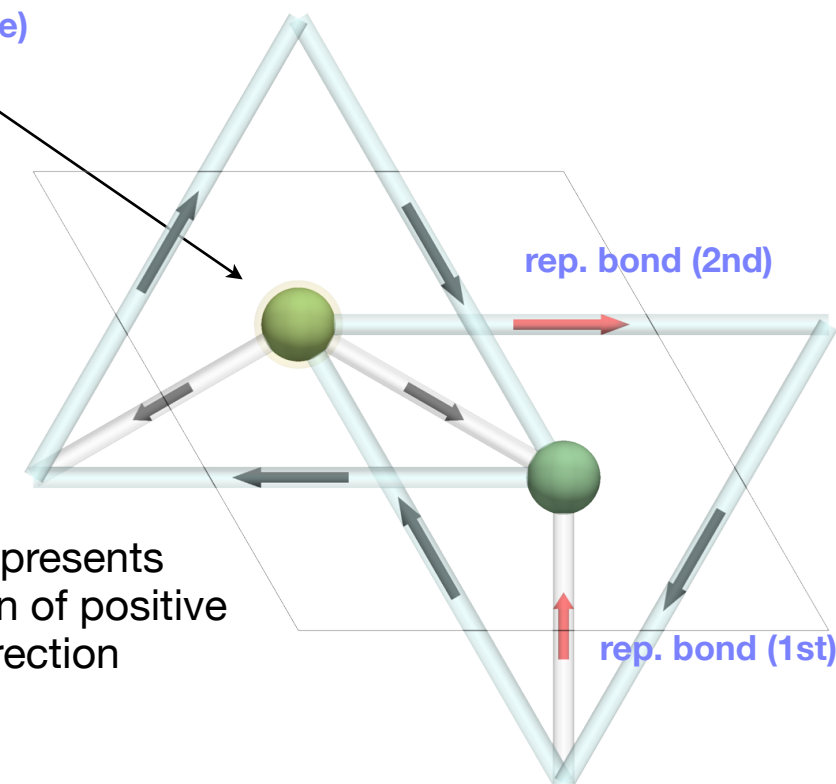


Table 3: dimension = 2 **basis of Hamiltonian matrix**

| # | orbital@atom(SL)    | # | orbital@atom(SL)    |
|---|---------------------|---|---------------------|
| 0 | $ p_z\rangle @C(1)$ | 1 | $ p_z\rangle @C(2)$ |

Table 4: Atomic basis (orbital part only)

| orbital       | definition |
|---------------|------------|
| $ p_x\rangle$ | $x$        |
| $ p_y\rangle$ | $y$        |
| $ p_z\rangle$ | $z$        |

**basis of atomic SAMB**

In the spinful case  
 $x \rightarrow (x, \uparrow), (x, \downarrow)$  etc.

# MultiPie - Model (Example : Graphene)

## Basis of site cluster

Table 7: 'C' (#1) site cluster (2c),  $-6m2$  Wyckoff (site) and site symmetry

| SL | position ( $\mathbf{s}$ )    | mapping                            |
|----|------------------------------|------------------------------------|
| 1  | [ 0.33333, 0.66667, 0.00000] | [1,2,3,10,11,12,16,17,18,19,20,21] |
| 2  | [ 0.66667, 0.33333, 0.00000] | [4,5,6,7,8,9,13,14,15,22,23,24]    |

fractional coordinate and mapping of symmetry op.

## Basis of bond cluster (1st neighbor)

Table 8: 1-th 'C'-'C' [1] (#1) bond cluster (3a@3f), ND,  $|\mathbf{v}| = 0.57735$  (cartesian) Wyckoff (bond@site) and directionality, distance of bond

| SL | vector ( $\mathbf{v}$ )      | center ( $\mathbf{c}$ )      | mapping                    | head  | tail  | $\mathbf{R}$ (primitive) |
|----|------------------------------|------------------------------|----------------------------|-------|-------|--------------------------|
| 1  | [ 0.33333, 0.66667, 0.00000] | [ 0.50000, 0.00000, 0.00000] | [1,-4,-8,11,-13,16,20,-23] | (2,1) | (1,1) | [0,-1,0]                 |
| 2  | [-0.66667,-0.33333, 0.00000] | [ 0.00000, 0.50000, 0.00000] | [2,-5,-7,10,-14,17,19,-22] | (2,1) | (1,1) | [1,0,0]                  |
| 3  | [ 0.33333,-0.33333, 0.00000] | [ 0.50000, 0.50000, 0.00000] | [3,-6,-9,12,-15,18,21,-24] | (2,1) | (1,1) | [0,0,0]                  |

fractional coordinate and mapping

(sublattice, +trans.)

$\mathbf{R} = \mathbf{R}_{\text{ket (tail)}} - \mathbf{R}_{\text{bra (head)}}$

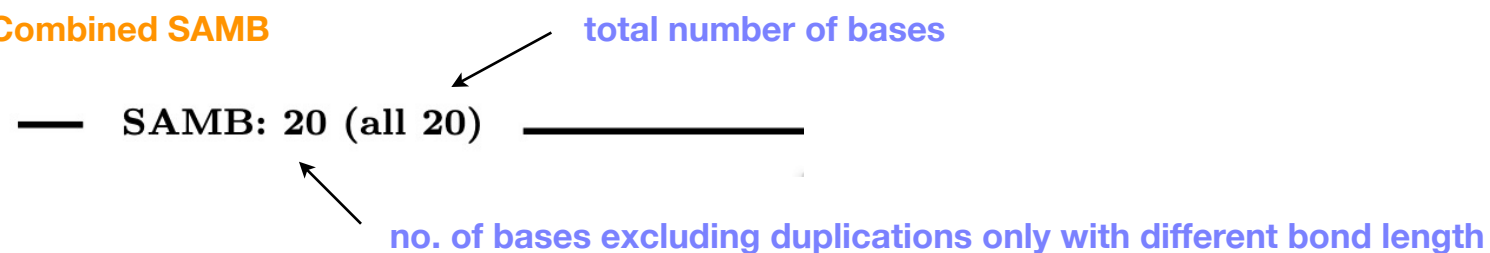
## Basis of bond cluster (2nd neighbor)

Table 9: 2-th 'C'-'C' [1] (#2) bond cluster (6b@6l), ND,  $|\mathbf{v}| = 1.0$  (cartesian)

| SL | vector ( $\mathbf{v}$ )      | center ( $\mathbf{c}$ )      | mapping        | head  | tail  | $\mathbf{R}$ (primitive) |
|----|------------------------------|------------------------------|----------------|-------|-------|--------------------------|
| 1  | [ 1.00000, 0.00000, 0.00000] | [ 0.83333, 0.66667, 0.00000] | [1,-11,16,-20] | (1,1) | (1,1) | [-1,0,0]                 |
| 2  | [ 0.00000, 1.00000, 0.00000] | [ 0.33333, 0.16667, 0.00000] | [2,-10,17,-19] | (1,1) | (1,1) | [0,-1,0]                 |
| 3  | [-1.00000,-1.00000, 0.00000] | [ 0.83333, 0.16667, 0.00000] | [3,-12,18,-21] | (1,1) | (1,1) | [1,1,0]                  |
| 4  | [-1.00000, 0.00000, 0.00000] | [ 0.16667, 0.33333, 0.00000] | [4,-8,13,-23]  | (2,1) | (2,1) | [1,0,0]                  |
| 5  | [ 0.00000,-1.00000, 0.00000] | [ 0.66667, 0.83333, 0.00000] | [5,-7,14,-22]  | (2,1) | (2,1) | [0,1,0]                  |
| 6  | [ 1.00000, 1.00000, 0.00000] | [ 0.16667, 0.83333, 0.00000] | [6,-9,15,-24]  | (2,1) | (2,1) | [-1,-1,0]                |

# MultiPie - Model (Example : Graphene)

## Combined SAMB



### • C : 'C' site-cluster name of site cluster

- \* bra:  $\langle p_z |$  basis of atomic SAMB
- \* ket:  $| p_z \rangle$
- \* wyckoff: 2c Wyckoff (site)

$$\boxed{z1} \quad Q_0^{(c)}(A_{1g}) = Q_0^{(a)}(A_{1g})Q_0^{(s)}(A_{1g})$$

expression of combined SAMB  
atomic x site-cluster

$$\boxed{z5} \quad Q_3^{(c)}(B_{1u}) = Q_0^{(a)}(A_{1g})Q_3^{(s)}(B_{1u})$$

## ID of basis

### • C;C\_001\_1 : 'C'-'C' bond-cluster name of bond cluster (tail; head\_neighbor\_multiplicity)

- \* bra:  $\langle p_z |$  basis of atomic SAMB
- \* ket:  $| p_z \rangle$
- \* wyckoff: 3a@3f Wyckoff (bond@site)

$$\boxed{z2} \quad Q_0^{(c)}(A_{1g}) = Q_0^{(a)}(A_{1g})Q_0^{(b)}(A_{1g})$$

expression of combined SAMB  
atomic x bond-cluster

$$\boxed{z6} \quad T_3^{(c)}(B_{1u}) = Q_0^{(a)}(A_{1g})T_3^{(b)}(B_{1u})$$

$$\boxed{z9} \quad T_{1,1}^{(c)}(E_{1u}) = \frac{\sqrt{2}Q_0^{(a)}(A_{1g})T_{1,1}^{(b)}(E_{1u})}{2}$$

$$\boxed{z10} \quad T_{1,2}^{(c)}(E_{1u}) = \frac{\sqrt{2}Q_0^{(a)}(A_{1g})T_{1,2}^{(b)}(E_{1u})}{2}$$

## matrix of atomic SAMB

- bra:  $\langle p_z |$  basis of atomic SAMB
- ket:  $| p_z \rangle$

$$\boxed{x1} \quad Q_0^{(a)}(A_{1g}) = \begin{bmatrix} 1 \end{bmatrix} \quad 1 \times 1 \text{ 行列}$$

## vector of site and bond cluster

\*\* Wyckoff: 2c Wyckoff (site)

$$\boxed{y1} \quad Q_0^{(s)}(A_{1g}) = \begin{bmatrix} \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \end{bmatrix} \quad 2\text{-site cluster}$$

$$\boxed{y2} \quad Q_3^{(s)}(B_{1u}) = \begin{bmatrix} \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2} \end{bmatrix}$$

\*\* Wyckoff: 3a@3f Wyckoff (bond@site)

$$\boxed{y3} \quad Q_0^{(b)}(A_{1g}) = \begin{bmatrix} \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3} \end{bmatrix} \quad 3\text{-bond cluster}$$

$$\boxed{y4} \quad T_3^{(b)}(B_{1u}) = \begin{bmatrix} \frac{\sqrt{3}i}{3}, \frac{\sqrt{3}i}{3}, \frac{\sqrt{3}i}{3} \end{bmatrix}$$

$$\boxed{y5} \quad T_{1,1}^{(b)}(E_{1u}) = \begin{bmatrix} 0, -\frac{\sqrt{2}i}{2}, \frac{\sqrt{2}i}{2} \end{bmatrix}$$

...



## Control file

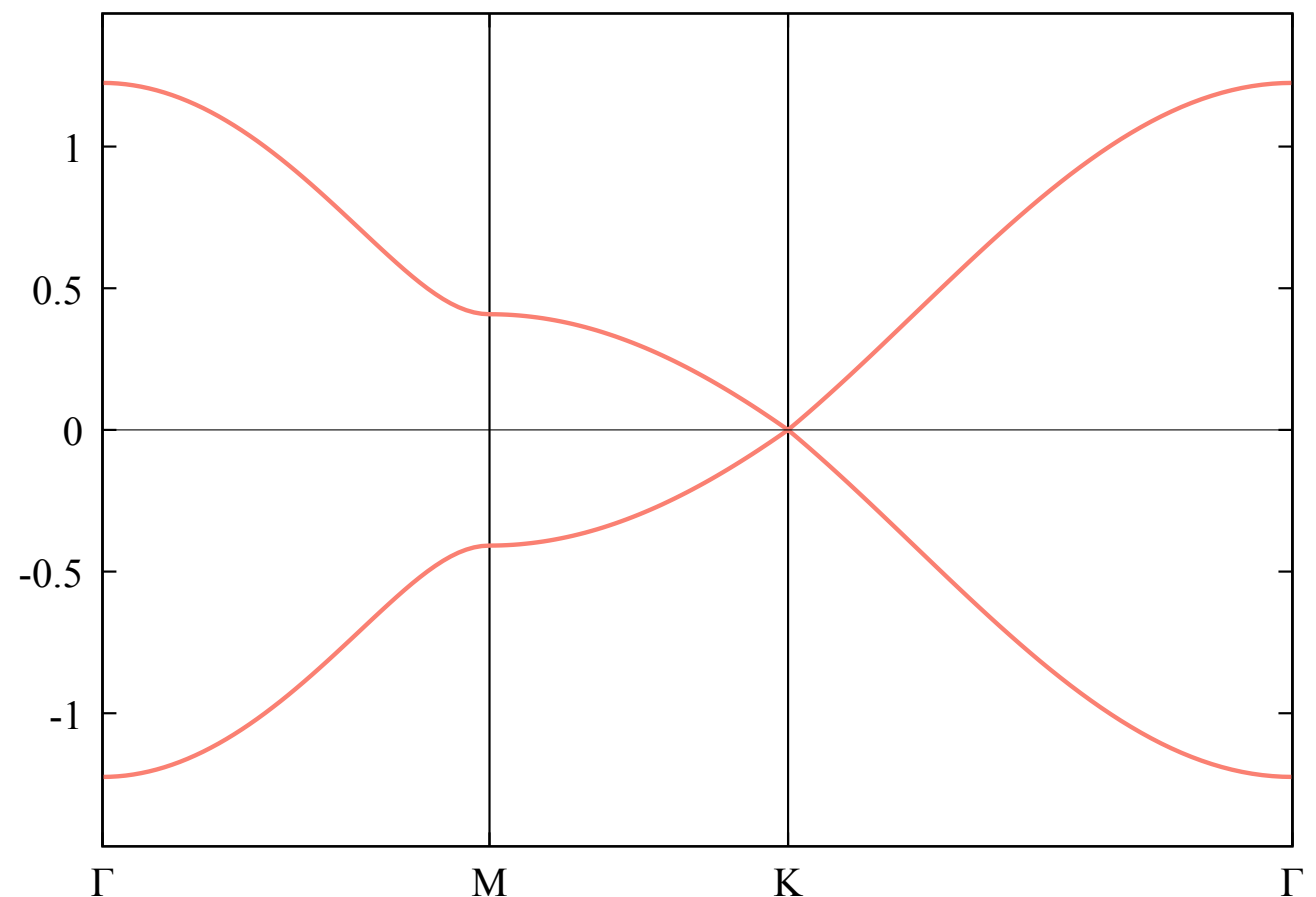
```
# graphene_ctrl.py
graphene_ctrl = {
    "samb": {
        "model": "graphene",    model name
        "parameter": {
            "z2": 1.0,        ID and weight of SAMBs used
        },
        "samb_figure": True,    generate QtDraw file for SAMB ?
    },
    "output": {
        "dispersion": {
            "k_point": {"Γ": "[0,0,0]", "K": "[1/3,1/3,0]", "M": "[1/2,0,0]"},    definition of k point
            "k_path": "Γ-M-K-Γ",    high-symmetry line ( connected with "-", discontinuities with "|" )
        },
    },
}
```

## Analyze & Output

(Now developing step by step)

```
$ mp_analyze ./graphene_ctrl.py
```

band dispersion [./graphene/output/graphene\\_dispersion.pdf](#)



# MultiPie - Matrix Element (Example : Graphene)

Matrix-element file `./graphene/graphene_matrix.py`

```
graphene = {
    "model": "graphene",
    "source": "graphene.pkl",
    "date": "2026-07-12 10:23:48",
    "select": {      filter conditions for selecting basis
        "X": ["Q", "G", "M", "T"],
        "l": [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11],
        ...
    },
    "dimension": 2,  dimension of matrix
    "ket_site": {
        "pz@C(1)": [0.3333333333333333, 0.6666666666666666, 0.0],
        "pz@C(2)": [0.6666666666666666, 0.3333333333333333, 0.0]
    },
    "index": {("C", 1, 1): (0, 1), ("C", 2, 1): (1, 1)},  (atom, sublattice, orbital rank) → (top of block matrix, size of block matrix)
    "vector": {      correspondence between cluster and bond vector (primitive cell)
        "C": [[0.0, 0.0, 0.0]],
        "C;C_001_1": [[0.33333333, 0.66666666, 0.0], [-0.66666666, -0.33333333, 0.0], [0.33333333, -0.33333333, 0.0]],
        ...
    },
    "cluster": {      correspondence between ID and cluster
        "z1": "C",
        "z2": "C;C_001_1",
        ...
    },
    "matrix": {      ID : { (n1, n2, n3, m, n) : matrix element, c, ... }
        "z1": {(0, 0, 0, 0, 0): "sqrt(2)/2", (0, 0, 0, 1, 1): "sqrt(2)/2"},
        "z2": {(0, -1, 0, 1, 0): "sqrt(6)/6", (1, 0, 0, 1, 0): "sqrt(6)/6", (0, 0, 0, 1, 0): "sqrt(6)/6",
                (0, 1, 0, 0, 1): "sqrt(6)/6", (-1, 0, 0, 0, 1): "sqrt(6)/6", (0, 0, 0, 0, 1): "sqrt(6)/6"},
        ...
    }
}
```

matrix element in momentum space

$$\mathbf{k} = 2\pi(\kappa_1 \mathbf{b}_1 + \kappa_2 \mathbf{b}_2 + \kappa_3 \mathbf{b}_3) \quad \text{fractional k vector (primitive cell)}$$

$$[\mathbb{Z}_j(\mathbf{k})]_{m,n} = \sum c(n_1, n_2, n_3, m, n) e^{2\pi i \mathbf{\kappa} \cdot (\mathbf{R} - \mathbf{r}_m + \mathbf{r}_n)}$$

$$\mathbf{R} = \mathbf{R}_n - \mathbf{R}_m = n_1 \mathbf{a}_1 + n_2 \mathbf{a}_2 + n_3 \mathbf{a}_3$$

(primitive cell)

frac. coord.  
in primitive unit cell

e.g.

```
position = list( graphene["ket_site"].values() )
r_m, r_n = position[m], position[n]
```



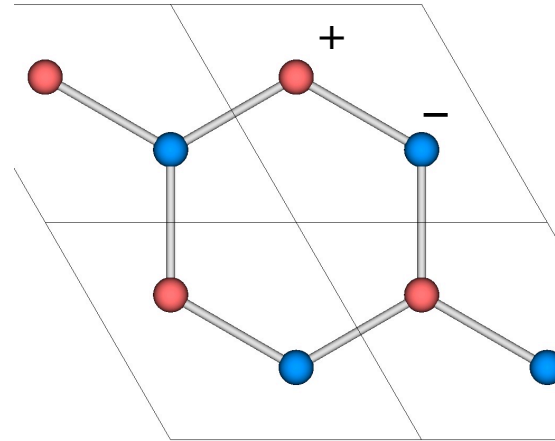
# MultiPie - How to use basis (Example : Graphene)

## Basis of non identity irreps.

mass term `samb/graphene_C.qtdw (y2)`

$$\boxed{\text{z5}} \quad Q_3^{(c)}(B_{1u}) = Q_0^{(a)}(A_{1g})Q_3^{(s)}(B_{1u})$$

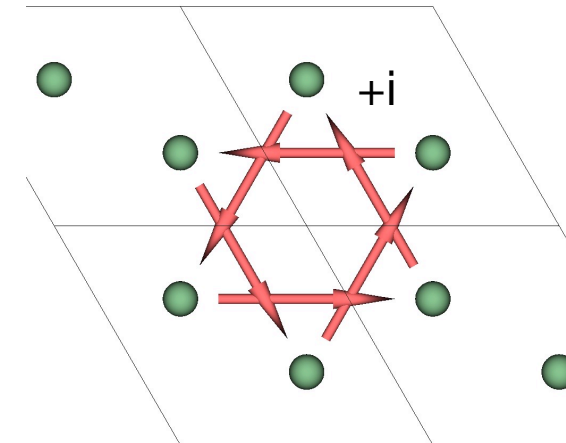
$$\boxed{\text{y2}} \quad Q_3^{(s)}(B_{1u}) = \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{bmatrix}$$



Haldane's magnetic flux `samb/graphene_C;C_002_1.qtdw (y10)`

$$\boxed{\text{z4}} \quad M_1^{(c)}(A_{2g}) = Q_0^{(a)}(A_{1g})M_1^{(b)}(A_{2g})$$

$$\boxed{\text{y10}} \quad M_1^{(b)}(A_{2g}) = \begin{bmatrix} \frac{\sqrt{6}i}{6} & \frac{\sqrt{6}i}{6} & \frac{\sqrt{6}i}{6} & \frac{\sqrt{6}i}{6} & \frac{\sqrt{6}i}{6} & \frac{\sqrt{6}i}{6} \end{bmatrix}$$



surface Rashba SOC if model is constructed with `spinful = True`, total number of bases becomes 80 (all 80)

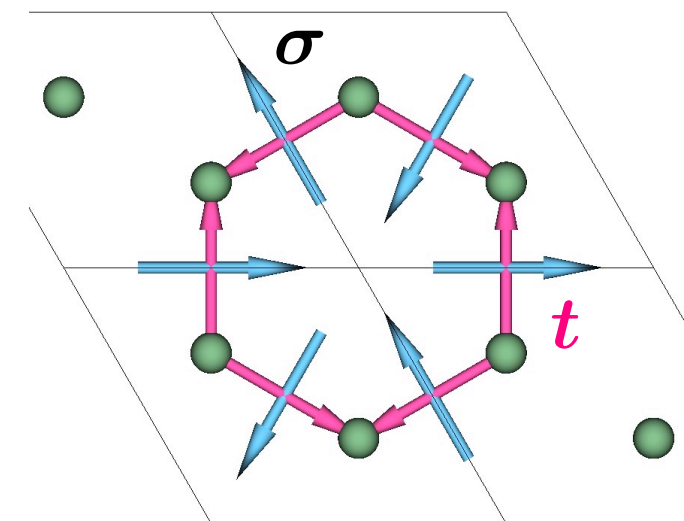
$$\boxed{\text{z12}} \quad Q_1^{(1,-1;c)}(A_{2u}) = \frac{\sqrt{2}M_{1,1}^{(1,-1;a)}(E_{1g})T_{1,2}^{(b)}(E_{1u})}{2} - \frac{\sqrt{2}M_{1,2}^{(1,-1;a)}(E_{1g})T_{1,1}^{(b)}(E_{1u})}{2}$$

$$\boxed{\text{x3}} \quad M_{1,1}^{(1,-1;a)}(E_{1g}) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & 0 \end{bmatrix}$$

$$\boxed{\text{x4}} \quad M_{1,2}^{(1,-1;a)}(E_{1g}) = \begin{bmatrix} 0 & -\frac{\sqrt{2}i}{2} \\ \frac{\sqrt{2}i}{2} & 0 \end{bmatrix}$$

$$\boxed{\text{y5}} \quad T_{1,1}^{(b)}(E_{1u}) = \begin{bmatrix} 0 & -\frac{\sqrt{2}i}{2} & \frac{\sqrt{2}i}{2} \end{bmatrix}$$

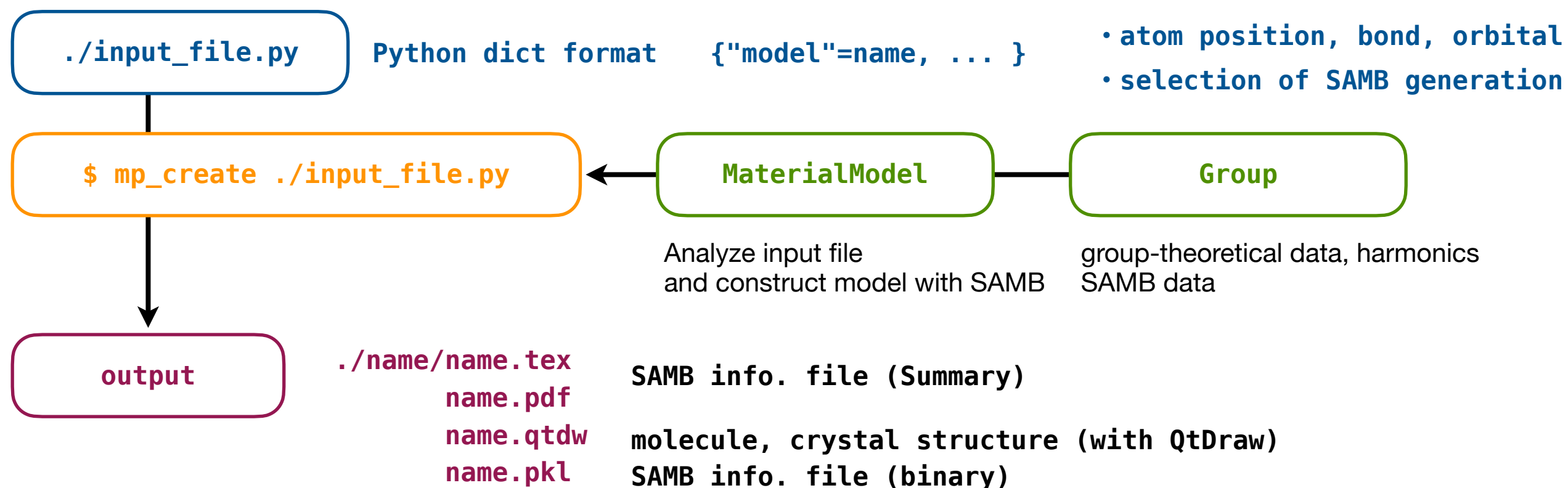
$$\boxed{\text{y6}} \quad T_{1,2}^{(b)}(E_{1u}) = \begin{bmatrix} \frac{\sqrt{6}i}{3} & -\frac{\sqrt{6}i}{6} & -\frac{\sqrt{6}i}{6} \end{bmatrix}$$



# **Model Generation and Physical-Quantity Calculations**

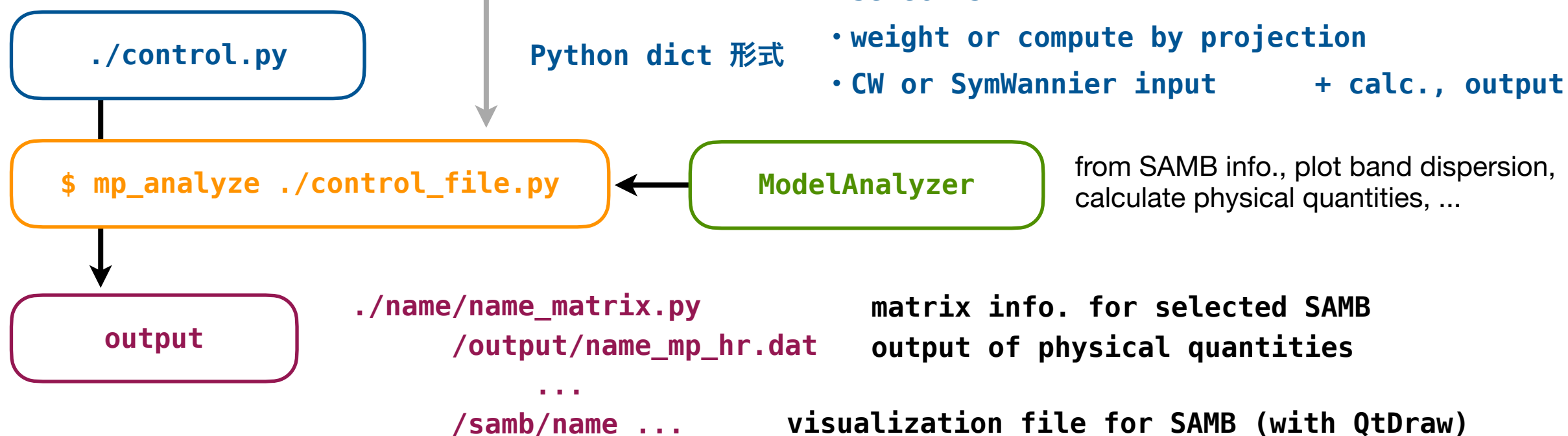
## 1. Model construction

```
from multipie import Group, MaterialModel
```



## 2. Plot band, physical quantities (Now developing step by step)

```
from multipie import ModelAnalyzer
```



# MultiPie - Input File (mp\_create)

## Input file

default values when omitted

```
default_model = {
  "model": "unknown",    model name : used as prefix for output folders/file names
  "group": "C1",         Crystallographic PG or SG name (Shoenflies symbol or SG no.)
  "cell": {},            unit-cell parameter (a, b, c, alpha, beta, gamma)
  "spinfo": False,       spinful orbital ?
  "site": {},            rep. site in unit cell  "name": ( [frac. coord.], [orbital list] )
  "bond": [],            rep. bond in unit cell  ( "tail", "head", [neighbor list] )
  "SAMB_select": { # select combined SAMB.          filter conditions in generating SAMB
    "X": ["Q", "G"], # type, "Q/G/T/M", []=all.
    "l": [], # rank, []=all.
    "Gamma": [], # "IR"=identity, []=all, "..."/["...", "..."]=specified irreps.
    "s": [], # spin, 0/1, []=all.
  },
  "atomic_select": { # select atomic SAMB.          filter conditions in generating atomic SAMB
    "X": [], # type, "Q/G/T/M", []=all.
    "l": [], # rank, []=all.
    "Gamma": [], # "IR"=identity, []=all, "..."/["...", "..."]=specified irreps.
    "s": [], # spin, 0/1, []=all.
  },
  "site_select": { # select site-cluster SAMB.      filter conditions in generating site-cluster SAMB
    "l": [], # rank, []=all.
    "Gamma": [], # "IR"=identity, []=all, "..."/["...", "..."]=specified irreps.
  },
  "bond_select": { # select bond-cluster SAMB.      filter conditions in generating bond-cluster SAMB
    "X": [], # type, "Q/T/M", []=all.
    "l": [], # rank, []=all.
    "Gamma": [], # "IR"=identity, []=all, "..."/["...", "..."]=specified irreps.
  },
  ...
}
```

# MultiPie - Input File (mp\_create)

```

...

"toroidal_priority": False,    in generating SAMB, priority for toroidal multipoes ?
"max_neighbor": 10,    # for DFT fitting, e.g. =200.    max. of neighbor in searching bond
"search_cell_range": (-2, 3, -2, 3, -2, 3),    max. range of cell in searching bond
    # (a1, a2, a3) range. for DFT fitting, e.g.=(-10, 10, -10, 10, -10, 10).
"qtdraw": {    option in generating QtDraw files
    "create": True,    # create QtDraw file ?    generate ?
    "mode": "standard",    # mode, "standard/detail".    standard or detail ?
    "view": None,    # [a,b,c], if None, default=[6,5,1].    default view
    #
    "rep_site": True,    # show representative site ?    highlight for rep. site ?
    "rep_bond": True,    # show representative bond ?    highlight for rep. bond ?
    #
    "max_neighbor": 5,    # max. neighbor to draw.    max. neighbor to plot bonds
    "cell_mode": None,    # "off/single/all", if None, single for SG, off for PG.    cell display mode
    "scale": None,    # default or magnification scale.    scale of site/bond size and width ?
    "site_radius": 0.07,    # base site radius.    default radius of site
    "bond_width": 0.02,    # base bond width.    default width of bond
},
"pdf": {    option in generating PDF
    "create": True,    # create PDF file ?    generate ?
    "common_samb": True,    # display common SAMB ?    display common bond SAMB ?
    "harmonics": True,    # display harmonics ?    display harmonics info. ?
    "max_neighbor": 5,    # max. neighbor to write (None: all).    max. of neighbor
},
}

```

# MultiPie - Control File (mp\_analyze)

## Control file

default values when omitted

```
default_control = {
    "samb": {
        "model": "unknown",    model name or None
        "select": {           select condition of SAMB      S=site name, R=orbital rank, N=neighbor bond
            # "site": [("A", [1,2])],    list of S or ( S, [R] ) format
            # "bond": [[0,1,2]],        list of S1;S2, [N], ( S1;S2, [N] ), ( S1;S2, R1;R2, [N] ) format
            # "X": [],                list of SAMB type, Q/G/T/M ( [] = all types)
            # "l": [],                list of SAMB rank, 0, 1, ..., 11 ( [] = all ranks)
            # "Gamma": ["A1g"],        list of SAMB irrep. ( "IR" = identity irrep., [] = all irreps.)
            # "s": [],                list of SAMB internal rank, 0, 1 ( [] = 0 and 1)
        },
        "parameter": {        finite weight (float or sympy const.) in case of empty dict. and given wannier, compute by projection
            # "z1": 1.0,                                     In case of file name, read "./name/filename"
        },
        "samb_figure": False,    output QtDraw files of SAMB ?
    },
    "wannier": {              Wannier info. from DFT → Wannier computation
        "cw": None,           file name
    },
    "output": {               info. for computing physical quantities
        "dir": "output",      output folder
        "fourier": { "tb_gauge": True },    use atomic position for phase (TB gauge) ?
        "dispersion": {      for band dispersion
            "k_point": {"Γ": [0,0,0]},    definition of k-point (primitive fractional)
            "k_path": "Γ",    high-symmetry line, connected with "-", discontinuities with "|"
            "local": [],        expectation values of local operators (Sx, Sy, Sz, Lx, Ly, Lz, Qu, Qv, Qyz, Qxz, Qxy)
            "z": [],            SAMB ID to compute expectation value
        },
        "dos": False,         compute DOS ?
    },
}
```



# Group Class

**Initialization** Generate crystallographic point group, space group, magnetic PG, and magnetic SG

**Group**(tag, with\_opt=False)

**tag**

example

|               |                            |                  |                       |
|---------------|----------------------------|------------------|-----------------------|
| • point group | "PG:no", "Schoenflies"     | no = 1 - 47      | "PG:5", "C3v"         |
| • space group | "SG:no", "Schoenflies", no | no = 1 - 230     | "SG:152", "D3^2", 152 |
| • magnetic PG | "MPG:HM_id"                | HM_id = PG.no.ID | "MPG:12.3.8"          |
| • magnetic SG | "MSG:BNS_id"               | BNS_id = SG.no   | "MSG:152.3"           |

**with\_opt**

load additional data ?

group.opt[key]

| dict key              | content                                             |
|-----------------------|-----------------------------------------------------|
| harmonics_multipole   | multipolar harmonics (dipole, quadrupole, octupole) |
| harmonics_irrep       | harmonics belonging to irrep. (irrep. → rank, no)   |
| representation_matrix | representation matrix                               |

**Global info.** Group.global\_info()

| dict key  | content                            | dict key        | content                              |
|-----------|------------------------------------|-----------------|--------------------------------------|
| id_set    | list of tag classified by category | harmonics       | max. of rank, name, definition, etc. |
| tag       | unique tag → conventional tag      | response_tensor | definition of tensor and multipole   |
| id        | conventional tag → unique tag      | root_cluster    | misc. of cluster SAMB                |
| character | alias and compatibility relation   |                 |                                      |

**Detail of API**

[https://cmt-mu.github.io/MultiPie/src/api\\_core/group.html](https://cmt-mu.github.io/MultiPie/src/api_core/group.html)

# MaterialModel Class

# MaterialModel - Overview

**Initialization** Create model from structure, symmetry, and atomic orbital info.

**MaterialModel**(**topdir**=None, **verbose**=False)

**topdir** top working directory (current dir when omitted)

**verbose** display log info. ?

**Dict data** [https://cmt-mu.github.io/MultiPie/src/api\\_core/material\\_model.html](https://cmt-mu.github.io/MultiPie/src/api_core/material_model.html)

**Property** **MaterialModel.group**

## Function

| function name                   | content                             | function name                   | content                        |
|---------------------------------|-------------------------------------|---------------------------------|--------------------------------|
| <b>x</b>                        | get atomic SAMB                     | <b>y</b>                        | get cluster SAMB               |
| <b>z</b>                        | get combined SAMB                   | <b>analyze</b>                  | analyze input data             |
| <b>cell_site_primitive</b>      | get site position in primitive cell | <b>clear</b>                    | clear info.                    |
| <b>get_atomic_samb</b>          | get all of atomic SAMB              | <b>get_cluster_samb</b>         | get all of cluster SAMB        |
| <b>get_combined_samb</b>        | get all of combined SAMB            | <b>get_combined_samb_matrix</b> | get combined SAMB matrix       |
| <b>get_hr</b>                   | get Hamiltonian matrix              | <b>get_ket_site</b>             | get ket and corresponding site |
| <b>get_multipole_expression</b> | get expression of multipole         | <b>get_samb_matrix</b>          | get full matrix info.          |
| <b>ket</b>                      | list of ket basis in LaTeX format   | <b>load</b>                     | load model file (binary)       |
| <b>save</b>                     | save model info. in binary          | <b>save_pdf</b>                 | save model info. in PDF        |
| <b>save_samb_hr</b>             | save SAMB matrix in hr format       | <b>save_samb_matrix</b>         | save SAMB matrix               |
| <b>save_samb_qtdraw</b>         | save QtDraw for SAMB                | <b>save_view</b>                | save structure file            |

## Detail of API

[https://cmt-mu.github.io/MultiPie/src/api\\_core/material\\_model.html](https://cmt-mu.github.io/MultiPie/src/api_core/material_model.html)

# ModelAnalyzer Class

# ModelAnalyzer - Overview

**Initialization** Compute and output physical quantities from model info.

**ModelAnalyzer**(**N1=50**, **N2=50**, **N3=50**, **topdir=None**, **verbose=False**)

**N1, N2, N3** no. of divisions in 1st Brillouin zone (primitive cell)

**topdir** top working directory (current dir when omitted)

**verbose** display log info. ?

## Property

| property name | content                         | property name  | content                   |
|---------------|---------------------------------|----------------|---------------------------|
| <b>HR</b>     | Hamiltonian matrix (real space) | <b>model</b>   | MaterialModel object      |
| <b>name</b>   | model name                      | <b>output</b>  | control (output related)  |
| <b>samb</b>   | control (samb related)          | <b>wannier</b> | control (wannier related) |

## Function

**analyze**(**control**) read control file, compute physical quantities, and output

**control** control file name or dict.

**set\_grid\_size**(**N1**, **N2**, **N3**) set no. of divisions

**N1, N2, N3** no. of divisions in 1st Brillouin zone (primitive cell)

## Detail of API

[https://cmt-mu.github.io/MultiPie/src/api\\_core/model\\_analyzer.html](https://cmt-mu.github.io/MultiPie/src/api_core/model_analyzer.html)